

Bearings, Scale Drawing and Constructions

From North, clockwise, three figures. Back bearing: $\pm 180^\circ$. Constructions: keep your arcs.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Bearings, Scale Drawing and Constructions

01 – Bearings and back bearings

bearing of A from B

b and $6b$

02 – Scale drawings and maps

1 cm represents 200 m

scale 1 : n

plot a point

03 – Constructions

perpendicular bisector

angle bisector

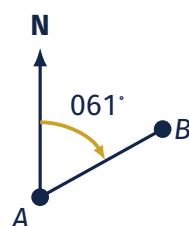
triangle from 3 sides

04 – Bearings with trigonometry

cosine rule

bearing of C from B

THREE RULES Every bearing obeys them:



1 from **North**

2 **clockwise**

3 **three figures**: 061° , not 61°

ZONE BACK BEARING

under 180° : add 180

$061^\circ \rightarrow 241^\circ$

ZONE BACK BEARING

over 180° : subtract 180

$330^\circ \rightarrow 150^\circ$

01 Bearings and Back Bearings

“The bearing of B from A ” — stand at A , face North, turn clockwise to B . The way back is always 180° different.

WHEN YOU SEE → YOU DO

“bearing of B from A ”

→ North line at A , clockwise to B

“bearing of A from B ” (back bearing)

→ $\pm 180^\circ$

“bearing of Y from X is b , of X from Y is $6b$ ”

→ $6b = b + 180$

an angle inside a triangle

→ use parallel North lines (co-interior 180°)

WORKED MODEL · PARALLEL NORTH LINES

The bearing of B from A is 070° ; $AB = CB$; angle $ABC = 50^\circ$. Find the bearing of C from A .

1 $BAC = \frac{180-50}{2} = 65^\circ$

isosceles triangle

ANSWER $070 + 65 = 135^\circ$

× **TRAP** Writing 61° . A bearing always has three figures: 061° .

IDEA 01

Practise: Bearings and Back Bearings

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Back bearing

CH 4 · TOPIC 3 2 marks

The bearing of B from A is 064° . Work out the bearing of A from B .

Q2 **DR ESLAM** Back bearing (big)

CH 4 · TOPIC 3 2 marks

The bearing of Q from P is 255° . Work out the bearing of P from Q .

SELECTED PRACTICE

Q3 **PRACTICE** Practice · Easy

CH 4 · TOPIC 3 2 marks

- 1 The bearing of a ship from a lighthouse is 050° .

Work out the bearing of the lighthouse from the ship.

(2 marks)

Q4**PRACTICE**

Practice · Easy

CH 4 · TOPIC 3 1 mark

7 The bearing of A from B is 310°

Circle the bearing of B from A .

050°	110°	130°	220°
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(1 mark)

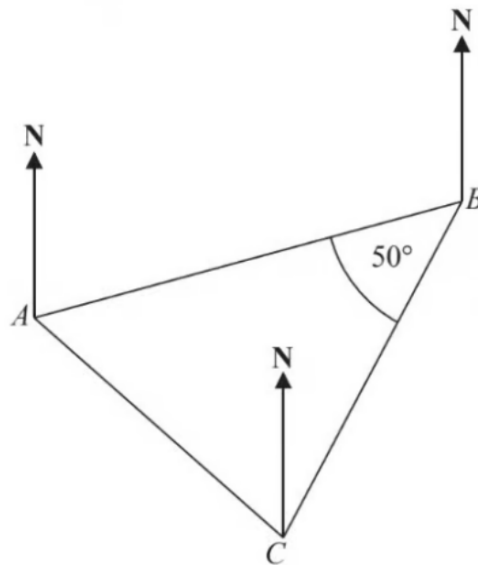
Q5**PRACTICE**

Practice · Medium

CH 4 · TOPIC 3

3 marks

- 5 The diagram shows the positions of three points, A , B and C , on a map.



The bearing of B from A is 070°

Angle ABC is 50°

$AB = CB$

Work out the bearing of C from A .

(3 marks)

Q6**PRACTICE**

Practice · Hard

CH 4 · TOPIC 3 6 marks

7 (a) A, B, C and D are four towns.

B is 25 kilometres due East of A.

C is 25 kilometres due North of A.

D is 45 kilometres due South of A.

North



C ×

A ×

D ×

×

B

Not to scale

Work out the bearing of B from C.

(2 marks)**(b)** Calculate the bearing of D from B.**(4 marks)**

PAST PAPERS

Q7**PAST PAPER**

Edexcel 4MA1 · May 2021 · Paper 2H · Q3

CH 4 · TOPIC 3 2 marks

- 3 The bearing of Paris from London is 149°

Work out the bearing of London from Paris.

○

(Total for Question 3 is 2 marks)

02 Scale Drawings and Maps

Map length $\times$ scale = real length. For $1 : n$, make both sides the same unit. To plot a point: protractor at the North line, measure, then the scaled distance.

WHEN YOU SEE $\rightarrow$ YOU DO

"1 cm represents 200 m"

$\rightarrow \times 200 \text{ m}; 1 : 20\,000$

"scale $1 : 87$ "

$\rightarrow \text{real} = \text{model} \times 87$

"mark the position of C"

$\rightarrow$ bearing from the North line, then scaled length

"C is on a bearing ... from A and ... from B"

$\rightarrow$ draw both lines; C is where they cross

WORKED MODEL · JANUARY 2022 P2H

Scale: 1 cm represents 200 m. C is 700 m from A on a bearing of 110° . Write the scale as $1 : n$.

1 $700 \div 200 = 3.5 \text{ cm on } 110^\circ$

plot C

2 $200 \text{ m} = 20\,000 \text{ cm}$

same units

ANSWER $1 : 20\,000$

IDEA 02

Practise: Scale Drawings and Maps

My questions first, then selected practice, then real past papers.

Q8 DR ESLAM Scale

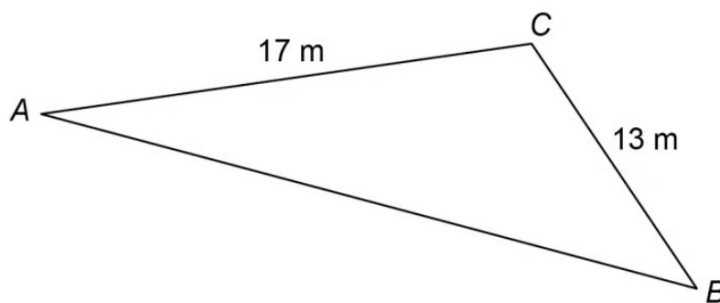
CH 4 • TOPIC 3 2 marks

On a map 1 cm represents 250 m. A path is 3.4 cm on the map. How long is the path? Write the scale as 1 : n .

SELECTED PRACTICE**Q9** PRACTICE Practice • Easy

CH 4 • TOPIC 3 1 mark

6 Here is a triangle.



Not drawn
accurately

Give a reason why the length of side AB **cannot** be 35 m

(1 mark)

Q10**PRACTICE**

Practice · Easy

CH 4 · TOPIC 3 2 marks

12 The scale of a map is 1 cm represents 25 m.

i) The length of a path is 240 m.

Work out the length, in centimetres, of the path on the map.

..... cm [1]

ii) The scale 1 cm represents 25 m can be written in the form 1: k .

Find the value of k .

$k =$ [1]

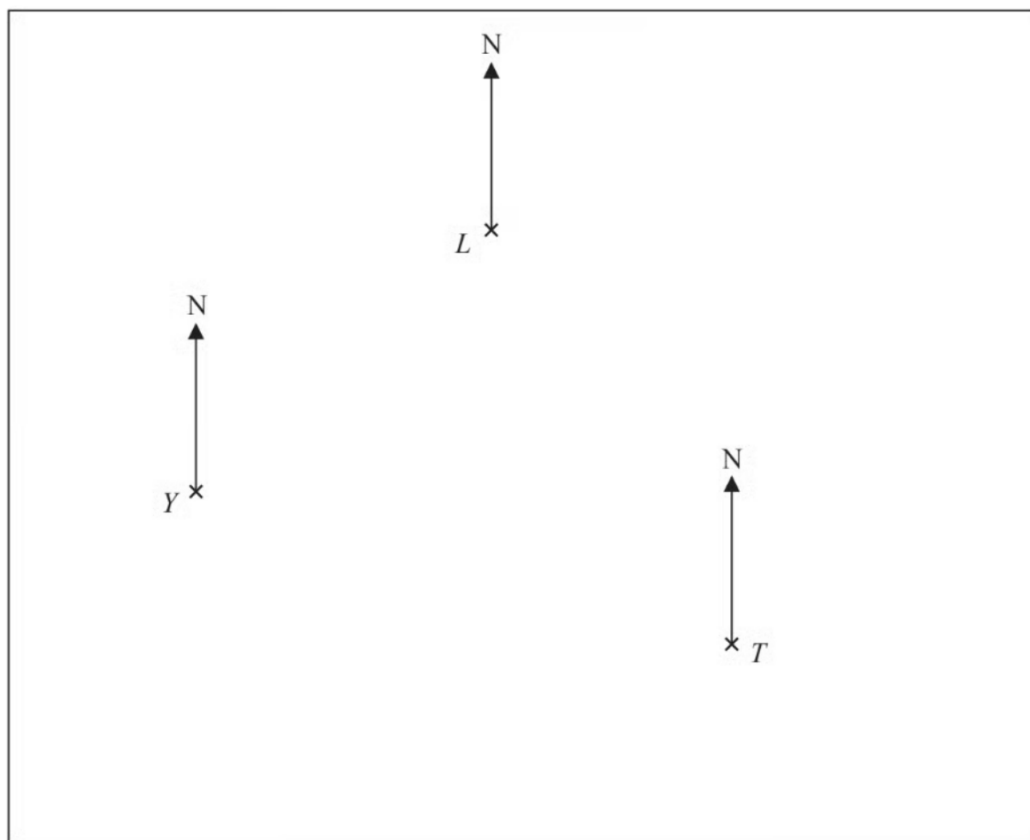
(2 marks)

Q11**PRACTICE**

Practice · Medium

CH 4 · TOPIC 3 3 marks

- 2 (a)** The diagram shows the positions of a lighthouse L , a yacht Y and a tanker T on a map.



Scale 1 cm represents 10 km

Measure the bearing of L from Y .

(1 mark)

- (b)** The tanker, T , sails 80km on a bearing of 320° .

Find the distance, in km, between the tanker and the lighthouse when the tanker is closest to the lighthouse.

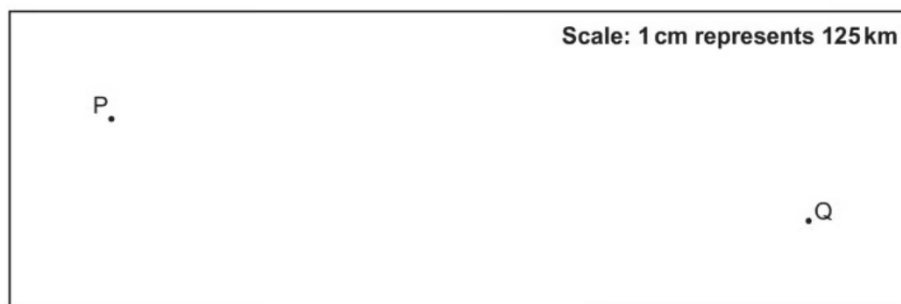
(2 marks)

Q12**PRACTICE**

Practice · Medium

CH 4 · TOPIC 3 6 marks

- 11 (a) The scale diagram below shows two cities, P and Q.



A plane departs from P at 09 47 and arrives at Q at 12 07.

Work out the average speed, in kilometres per hour, of the plane.

..... km/h

(5 marks)

- (b) Give one reason why your answer may be inaccurate.

(1 mark)

PAST PAPERS

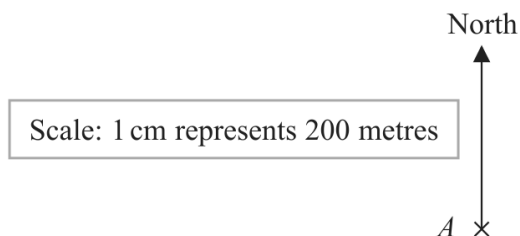
Q13

PAST PAPER

Edexcel 4MA1 · January 2022 · Paper 2H · Q5

CH 4 · TOPIC 3 4 marks

- 5 The scale diagram shows the position on a map of a house, A



House C is on a bearing of 110° from A
The distance from A to C is 700 m

- (a) Mark the position of C on the diagram with a cross ($\times$)
Label your cross C
- (b) Write the scale of the map in the form $1 : n$

1 :

(Total for Question 5 is 4 marks)

03 Constructions

Compasses fixed, arcs from each end (or each arm). Never rub the arcs out — they are the marks.

WHEN YOU SEE → YOU DO

“perpendicular bisector of AB ”

→ same-radius arcs from A and B , join the crossings

“bisector of angle ABC ”

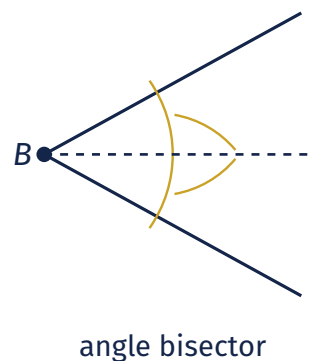
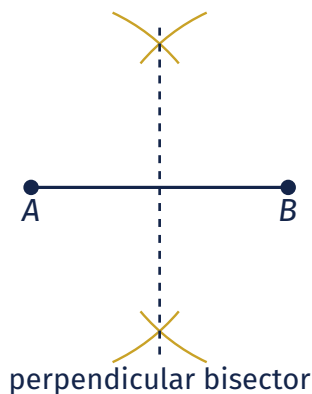
→ arc from B on both arms, then arcs from those points

“construct an equilateral triangle / triangle with sides ...”

→ base, then arcs of the other two lengths

“show all your construction lines”

→ keep every arc



IDEA 03

Practise: Constructions

My questions first, then selected practice, then real past papers.

SELECTED PRACTICE**Q14**

PRACTICE

Practice • Easy

CH 4 • TOPIC 3

3 marks

- 9 An equilateral triangle has side length 16 metres.

Using ruler and compasses only, construct a scale drawing of the triangle.

Use the scale 1 centimetre represents 2 metres.

Scale: 1 cm represents 2 m

(3 marks)

Q15

PRACTICE

Practice • Medium

CH 4 • TOPIC 3

2 marks

4

A ————— B

Use ruler and compasses to **construct** the perpendicular bisector of the line segment AB .

You must show all your construction lines.

(2 marks)

Q16**PRACTICE**

Practice · Medium

CH 4 · TOPIC 3 2 marks

- 10 Use ruler and compasses to construct the perpendicular bisector of the line DE . You must show all your construction lines.

D  E

(2 marks)

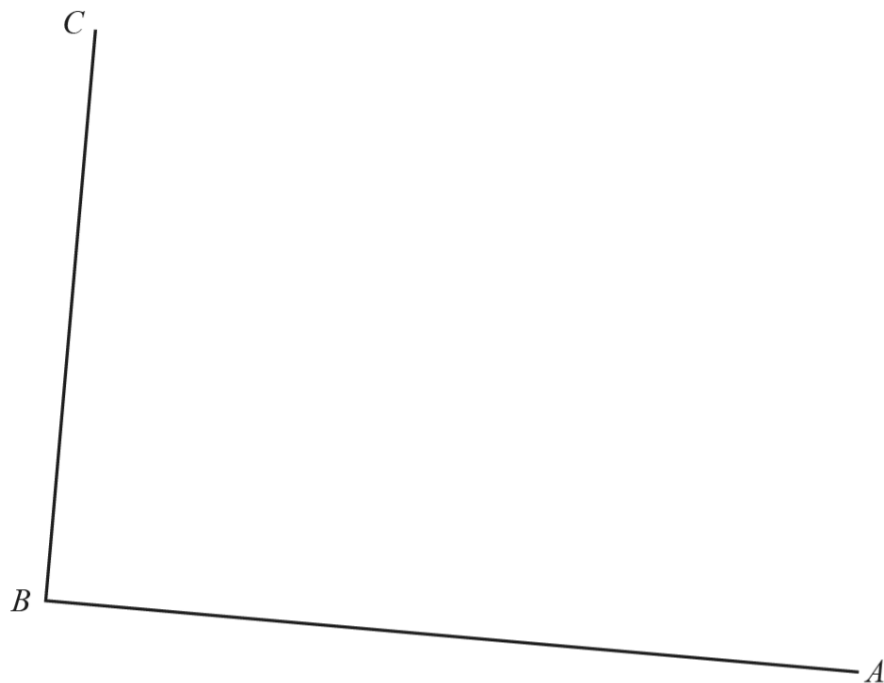
Q17**PRACTICE**

Practice · Hard

CH 4 · TOPIC 3

3 marks

- 8 The scale drawing shows two boundaries, AB and BC , of a field $ABCD$. The scale of the drawing is 1 cm represents 8 m.



Scale: 1 cm to 8 m

The boundaries CD and AD of the field are each 72 m long.

- i) Work out the length of CD and AD on the scale drawing.

..... cm [1]

- ii) **Using a ruler and compasses only**, complete accurately the scale drawing of the field.

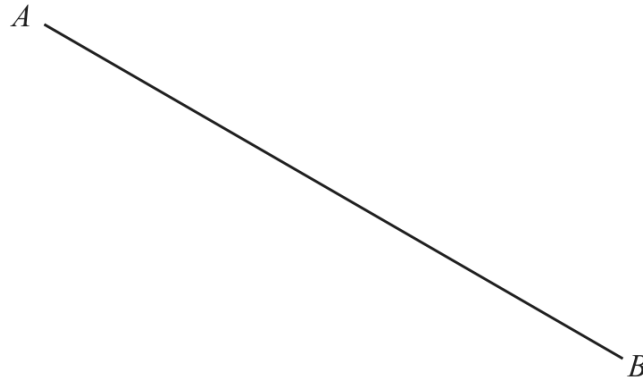
[2]

(3 marks)

PAST PAPERS

Q18 **PAST PAPER** Edexcel 4MA1 · January 2020 · Paper 2HR · Q6**CH 4 · TOPIC 3** **2 marks**

- 6 Use ruler and compasses only to construct the perpendicular bisector of the line AB .
You must show all your construction lines.



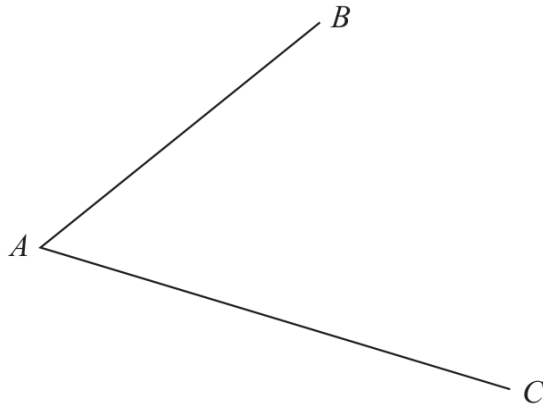
(Total for Question 6 is 2 marks)

Q19**PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 2H · Q5

CH 4 · TOPIC 3 2 marks

- 5 Use ruler and compasses to construct the bisector of angle BAC .
You must show all your construction lines.



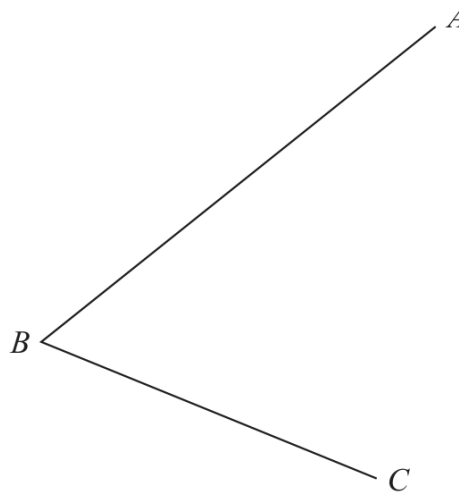
(Total for Question 5 is 2 marks)

Q20**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q2

CH 4 • TOPIC 3 2 marks

- 2 Using ruler and compasses only, construct the bisector of angle ABC .
You must show all your construction lines.



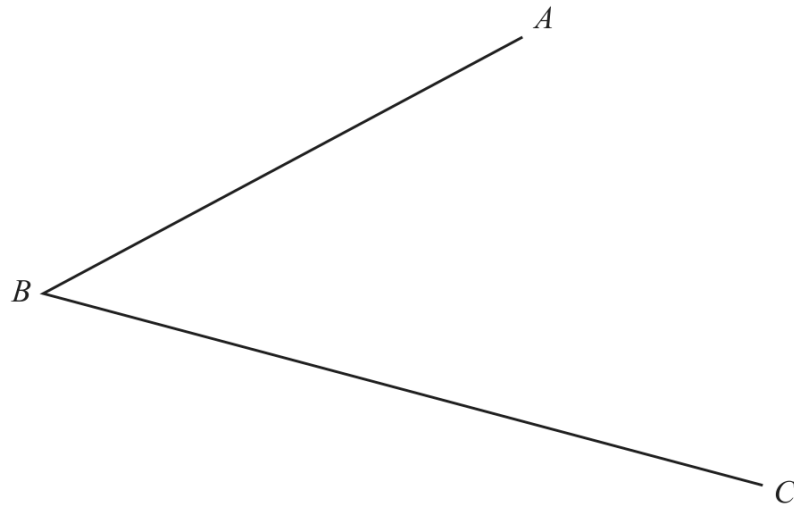
(Total for Question 2 is 2 marks)

Q21**PAST PAPER**

Edexcel 4MA1 · November 2024 · Paper 1H · Q2

CH 4 · TOPIC 3 2 marks

- 2 Use ruler and compasses only to construct the bisector of angle ABC
You must show all your construction lines.



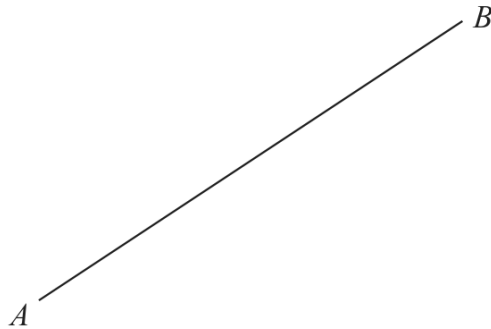
(Total for Question 2 is 2 marks)

Q22**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 2H · Q4

CH 4 · TOPIC 3 2 marks

- 4 Using ruler and compasses only, construct the perpendicular bisector of the line AB
Show all your construction lines.



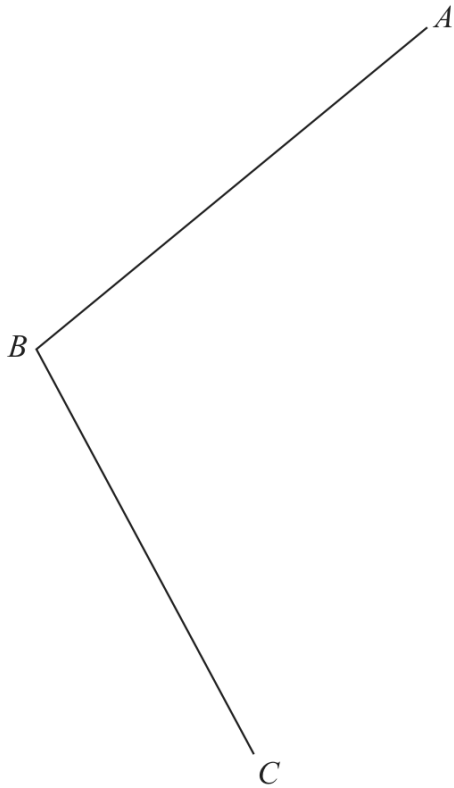
(Total for Question 4 is 2 marks)

Q23**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 1H • Q3

CH 4 • TOPIC 3 2 marks

- 3 Using ruler and compasses only, construct the bisector of angle ABC
Show all your construction lines.



(Total for Question 3 is 2 marks)

04 Bearings with Trigonometry

Put the bearings into a triangle using parallel North lines. Then the cosine or sine rule finds a side or an angle, and the angle turns back into a bearing.

WHEN YOU SEE → YOU DO

two sides and the angle between

→ cosine rule for the third side

“calculate the bearing of C from B”

→ find the angle at B, then add to/subtract from a known bearing

“total time for the walk”

→ find the missing side, total distance ÷ speed

a right angle (due east / due north)

→ SOH CAH TOA

Sketch with North lines

Angle inside the triangle

Cosine / sine rule

Back to a bearing

WORKED MODEL • NOVEMBER 2023 P1H

B is 3.6 km from A on 054° ; C is 8.4 km from B on 132° . Walking at 6 km/h, total time for $A \rightarrow B \rightarrow C \rightarrow A$?

1 angle $ABC = 54 + (180 - 132) = 102^\circ$

co-interior North lines

2 $AC^2 = 3.6^2 + 8.4^2 - 2(3.6)(8.4)\cos 102^\circ \Rightarrow AC = 9.80$

cosine rule

ANSWER $\frac{21.8}{6} = 3.63 \text{ h} = 3 \text{ h } 38 \text{ min}$

IDEA 04

Practise: Bearings with Trigonometry

My questions first, then selected practice, then real past papers.

Q24**DR ESLAM**

Coordinates

CH 4 · TOPIC 3

4 marks

B is 6 km from A on a bearing of 090° ; C is 8 km from B on a bearing of 180° .
Find the bearing of C from A .

SELECTED PRACTICE

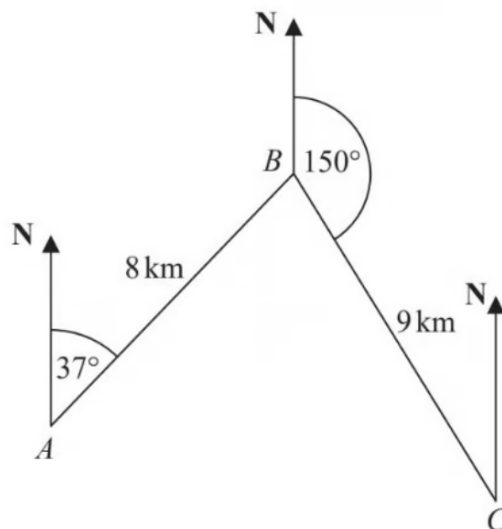
Q25

PRACTICE

Practice · Hard

CH 4 · TOPIC 3 5 marks

- 1 The diagram shows the positions of three towns, Acton (A), Barston (B) and Chorlton (C).



Barston is 8 km from Acton on a bearing of 037° .
Chorlton is 9 km from Barston on a bearing of 150° .

Find the bearing of Chorlton from Acton.
Give your answer correct to 1 decimal place.
You must show all your working.

(5 marks)

Q26**PRACTICE**

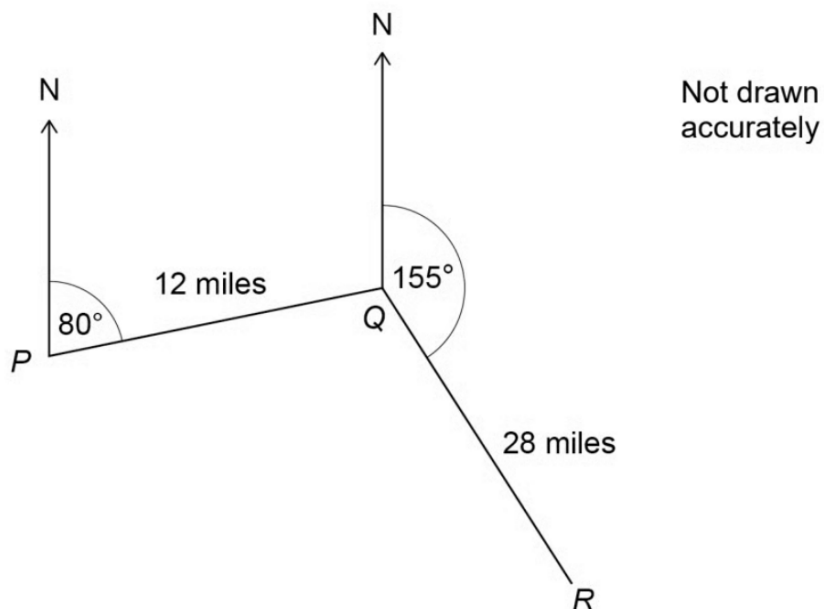
Practice · Hard

CH 4 · TOPIC 3 4 marks

- 5 A ship sails from P to Q and then from Q to R .

Q is 12 miles from P , on a bearing of 080°

R is 28 miles from Q , on a bearing of 155°



Work out the direct distance from P to R .

(4 marks)

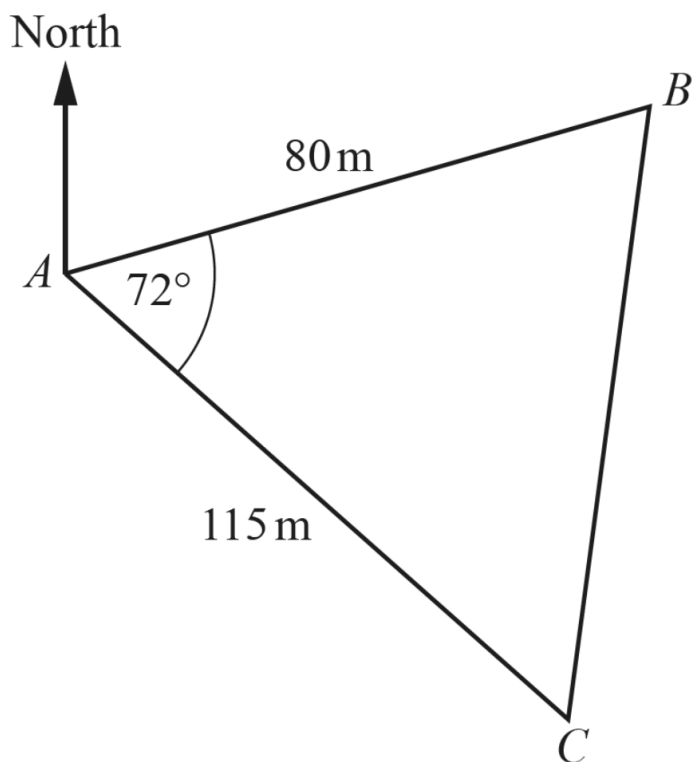
Q27**PRACTICE**

Practice · Hard

CH 4 · TOPIC 3

11 marks

9 (a)

NOT TO
SCALE

The diagram shows the positions of three points A , B and C in a field.

Show that BC is 118.1 m, correct to 1 decimal place.

(3 marks)

(b) Calculate angle ABC .Angle $ABC = \dots\dots\dots$

(3 marks)

(c) The bearing of C from A is 147° . Find the bearing ofi) A from B ,

[3]

ii) B from C .

[2]

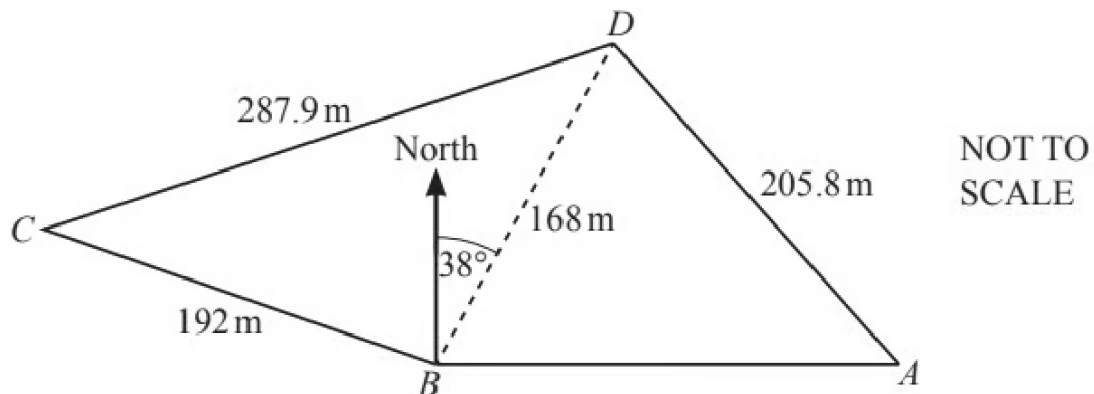
(5 marks)

Q28**PRACTICE**

Practice · Hard

CH 4 · TOPIC 3 6 marks

11



The diagram shows a field, $ABCD$, on horizontal ground. $BC = 192$ m, $CD = 287.9$ m, $BD = 168$ m and $AD = 205.8$ m.

Angle $CBD = 106^\circ$.

i) The bearing of D from B is 038° .

Find the bearing of C from B .

[1]

ii) A is **due east** of B .

Calculate the bearing of D from A .

[5]

(6 marks)

PAST PAPERS

Q29

PAST PAPER

Edexcel 4MA1 · January 2021 · Paper 1H · Q16

CH 4 · TOPIC 3 5 marks

16 The diagram shows the positions of three ships, A , B and C .

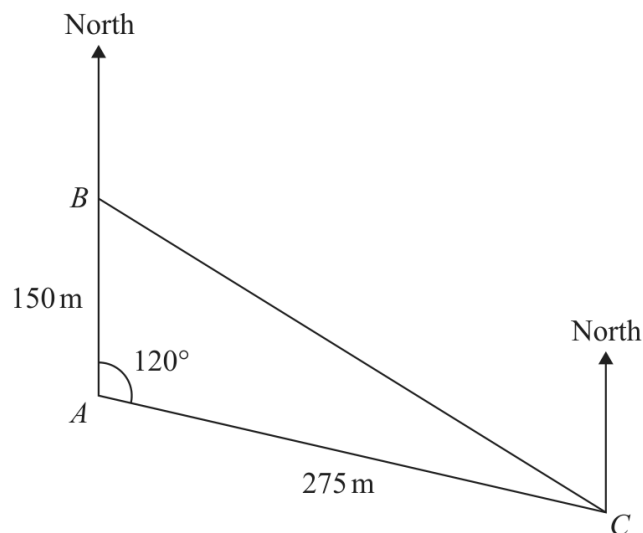


Diagram **NOT**
accurately drawn

Ship B is due north of ship A .

The bearing of ship C from ship A is 120°

Calculate the bearing of ship C from ship B .

Give your answer correct to the nearest degree.

o

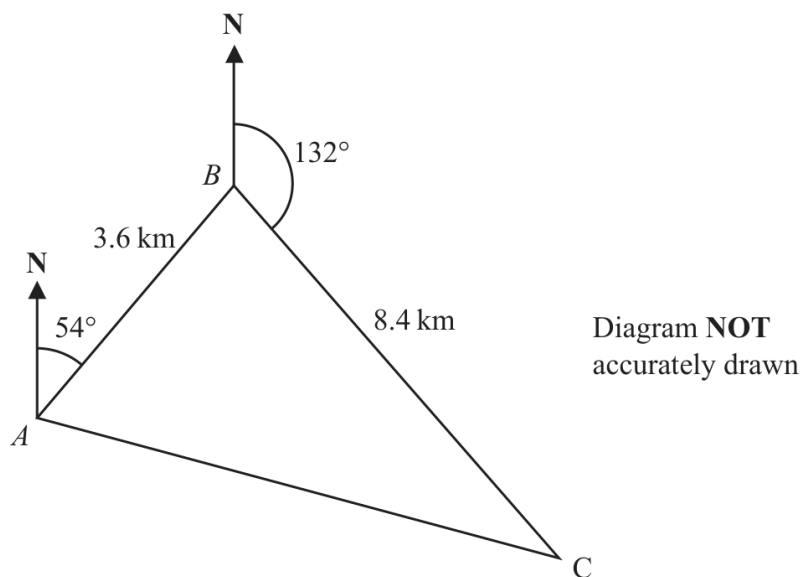
(Total for Question 16 is 5 marks)

Q30**PAST PAPER**

Edexcel 4MA1 · November 2023 · Paper 1H · Q18

CH 4 · TOPIC 3 5 marks

18 The diagram shows the positions of three villages, A , B and C



The bearing of B from A is 054°

The bearing of C from B is 132°

Melur walks from A to B

She then walks from B to C and from C to A

Melur walks at an average speed of 6 km/h

Work out the total time Melur takes.

Give your answer in hours and minutes.

..... hours minutes

(Total for Question 18 is 5 marks)

Q31**PAST PAPER**

Edexcel 4MA1 · November 2024 · Paper 2H · Q24

CH 4 · TOPIC 3 6 marks

24 A , B and C are three points on horizontal ground.

$$AB = 8.4 \text{ metres } BC = 9.2 \text{ metres}$$

B is on a bearing of 067° from A

C is on a bearing of 129° from B

Calculate the bearing of A from C

Give your answer correct to the nearest degree.

o

(Total for Question 24 is 6 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | |
|-------------------------|--|
| 01 Bearings | From North, clockwise, three figures; back bearing ± 180 . |
| 02 Scale | Map $\times$ scale = real; $1 : n$ in the same units. |
| 03 Constructions | Arcs from both ends / both arms; keep the arcs. |
| 04 With trig | North lines $\rightarrow$ triangle $\rightarrow$ cosine/sine rule $\rightarrow$ bearing. |

BOTTOM LINE Draw the North line at the point you stand on $\rightarrow$ turn clockwise $\rightarrow$ three figures.

Answers

Bearings, Scale Drawing and Constructions • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • BEARINGS AND BACK BEARINGS

Q1	Dr Eslam • Back bearing	$064 + 180 = 244^\circ$
Q2	Dr Eslam • Back bearing (big)	$255 - 180 = 075^\circ$
Q3	Practice • Easy	230°
Q4	Practice • Easy	130°
Q5	Practice • Medium	$BAC = 65^\circ: 135^\circ$
Q6	Practice • Hard	(a) 135° (b) 209°
Q7	Edexcel 4MA1 • May 2021 • Paper 2H • Q3	329°

IDEA 02 • SCALE DRAWINGS AND MAPS

Q8	Dr Eslam • Scale	850 m; 1 : 25 000
Q9	Practice • Easy	$17 + 13 = 30 < 35$: the triangle could not close
Q10	Practice • Easy	(i) 9.6 cm (ii) $k = 2500$
Q11	Practice • Medium	(a) measure (about 045°) (b) the shortest distance is perpendicular to the path
Q12	Practice • Medium	(a) distance $\div \frac{7}{3} h$ (b) the plane does not fly in a straight line / constant speed
Q13	Edexcel 4MA1 • January 2022 • Paper 2H • Q5	(a) 3.5 cm on 110° (b) 1 : 20 000

IDEA 03 • CONSTRUCTIONS

- Q14** Practice • Easy equilateral triangle of side 8 cm with arcs
- Q15** Practice • Medium arcs above and below, line through the crossings
- Q16** Practice • Medium perpendicular bisector of DE with arcs
- Q17** Practice • Hard (i) 9 cm (ii) arcs of 9 cm from C and from A meet at D
- Q18** Edexcel 4MA1 • January 2020 • Paper 2HR • Q6 perpendicular bisector with arcs
- Q19** Edexcel 4MA1 • January 2020 • Paper 2H • Q5 angle bisector with arcs
- Q20** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q2 angle bisector with arcs
- Q21** Edexcel 4MA1 • November 2024 • Paper 1H • Q2 angle bisector with arcs
- Q22** Edexcel 4MA1 • June 2025 • Paper 2H • Q4 perpendicular bisector with arcs
- Q23** Edexcel 4MA1 • November 2025 • Paper 1H • Q3 angle bisector with arcs

IDEA 04 • BEARINGS WITH TRIGONOMETRY

- Q24** Dr Eslam · Coordinates $\tan^{-1} \frac{6}{8} = 36.9^\circ$; $180 - 36.9 = 143.1^\circ$
- Q25** Practice · Hard 098.6°
- Q26** Practice · Hard 33.2 miles
- Q27** Practice · Hard (a) cosine rule: 118.1 m (b) 67.9° (c)(i) 255° (ii) 007°
- Q28** Practice · Hard (i) 292° (ii) 310°
- Q29** Edexcel 4MA1 · January 2021 · Paper 1H · Q16 140°
- Q30** Edexcel 4MA1 · November 2023 · Paper 1H · Q18 $AC = 9.80$ km; 21.8 km at 6 km/h: 3 hours 38 minutes
- Q31** Edexcel 4MA1 · November 2024 · Paper 2H · Q24 280°